Buck Converter Design

Specifications:

$V_{in} = 12V$, range 10-14 V
$V_{out} = 5V$
$W_{out} = 20W$
$I_{load} = 4A$
$f_{sw} = 100kHz$
$\Delta I_L = 1A$ (25% of I_{load})
$\Delta V_{out} = 50mV$ (1% of V_{out})

Calculation of L, C, duty cycle range and CCM boundary

Using worst case $V_{in} = 14V$ for sizing of the inductor to achieve desired current ripple for full range of input voltage.

$$L = \frac{V_{out}}{\Delta I_L \, f_{sw}} \left(1 - \frac{V_{out}}{V_{in}}\right)$$

$L = 3.214 \times 10^{-5} H$

Picking $33\mu H$ is a safe choice.

Checking the ripple with $V_{in} = 10V$:

$$\Delta I_L = \frac{V_{out}}{L \, f_{sw}} \left(1 - \frac{V_{out}}{V_{in}}\right)$$

$\Delta I_L = 0.758A$

Duty cycle D for the range of V_{in} :

$$D = \frac{t_{on}}{T} = \frac{V_{out}}{V_{in}}$$

$D = 0.5$ for $V_{in} = 10V$
$D = 0.417$ for $V_{in} = 12V$
$D = 0.357$ for $V_{in} = 14V$

Capacitor sizing:

$$C = \frac{\Delta I_L}{8 \, f_{sw} \, \Delta V_{out}}$$

$C = 2.5 \times 10^{-5} C = 25\mu F$

Considering ESR effects, **reasonable pick is $C = 47\mu F$ with $ESR = 10m\Omega$,**
giving roughly 27 mV capacitive and 10 mV ESR ripple, leaving space below 50 mV.

$$\Delta V_{ESR} = \Delta I_L \cdot ESR$$

CCM Operation:

CCM boundary is reached when $I_{Lmin} = 0$

$$I_{Lmin} = I_{load} - \Delta I_L/2$$

$$I_{load,min} = \frac{\Delta I_L}{2}$$

The worst case is at Vin = 14V and $I_{load} = 0.5A$.

Thus, the converter enters DCM below 2.5W.

Open-loop power stage design in MATLAB Simulink

PWM Generator is built using a Sawtooth Generator (100kHz frequency) with a comparator and a constant duty value of D = 0.417 (For 12V input voltage).

For switching, I use a MOSFET with threshold voltage of 2V and drain-source on resistance of 0.01Ω
Gain of 3 is required to trigger MOSFET's threshold.

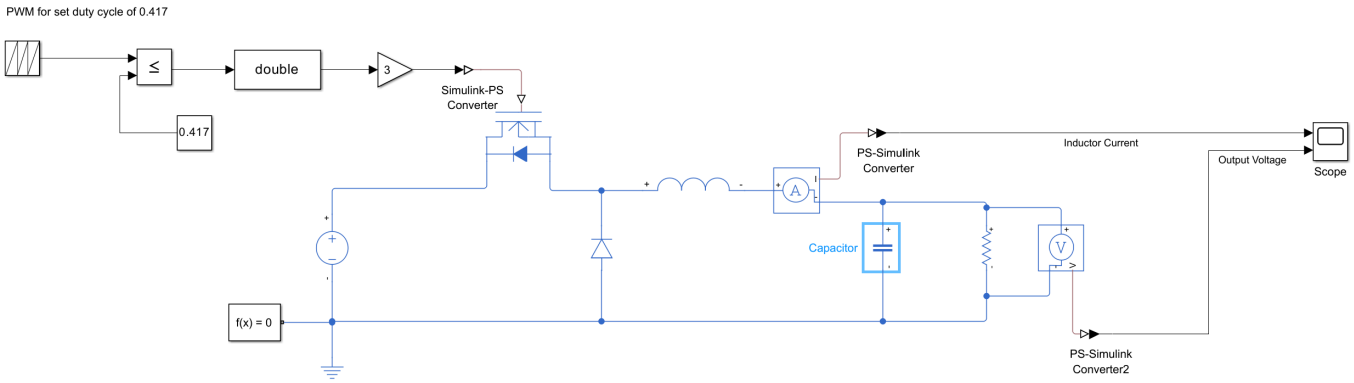


Figure 1. Open-loop power stage Simulink model.

Using standard diode from the Simscape Electrical library with forward voltage of 0.6V and on-resistance of 0.3Ω gives a large loss on that diode, bringing output voltage to about 4.07V (in simulation) instead of desirable 5V.

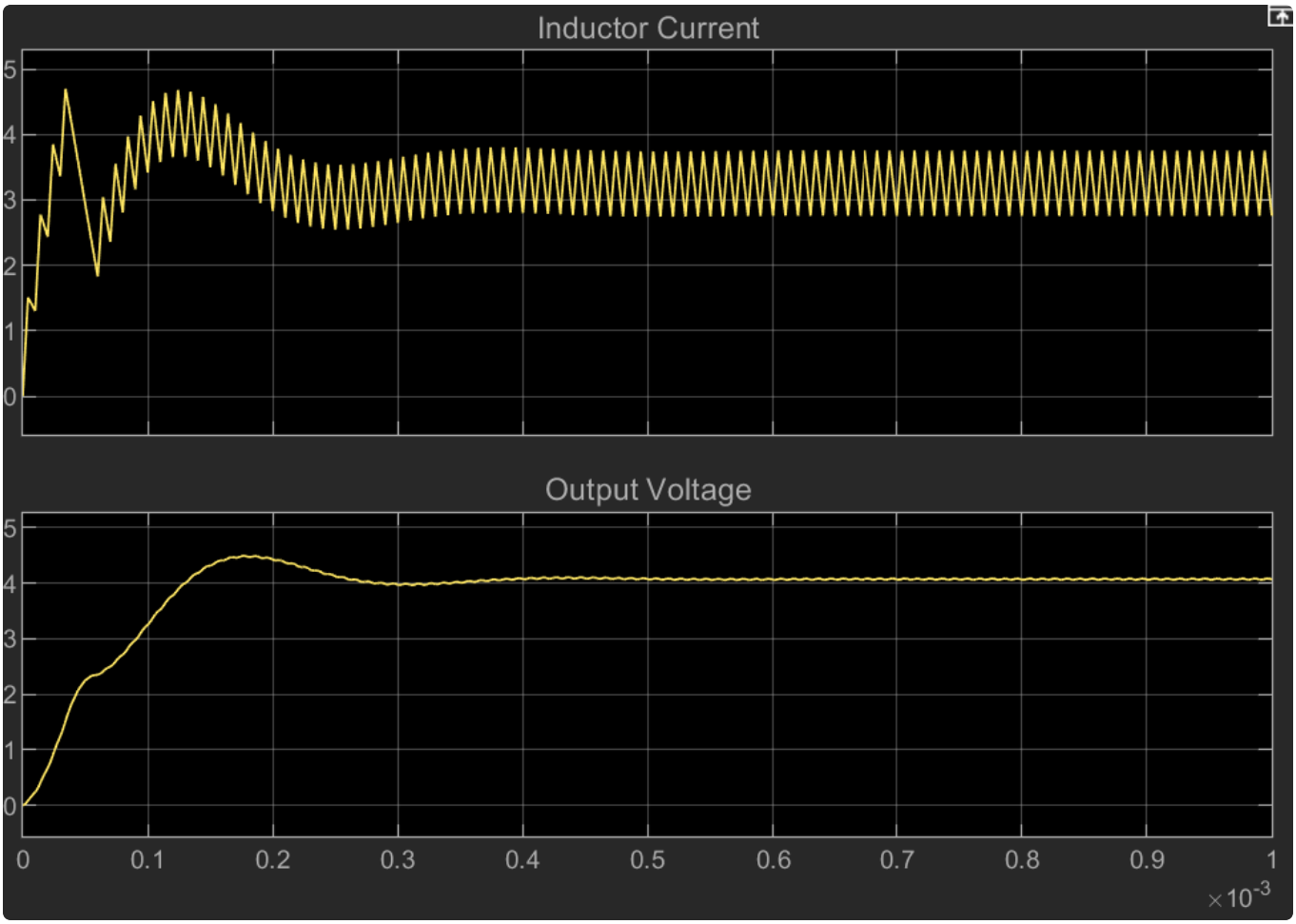


Figure 2. Inductor Current (A) and Output Voltage (V) readings for the standard diode model in open-loop stage.

Let me account for that diode in calculation to check whether the simulation is accurate.

Volt-second balance equation:

$$(V_{in} - V_{out})D + (-V_{out} - V_f - R_{on}I_{load})(1 - D) = 0$$

$$V_{out} = DV_{in} - (1 - D)(V_f + R_{on}I_{load})$$

Measured I_{load} is approximately 3.26A.
Solving for V_{out} yields:

$V_{out} = 4.08V$

Which almost perfectly matches with the simulation.

Now let's be more realistic and use a Schottky Diode with a forward voltage of 0.35V and on resistance of 30 $m\Omega$ (e.g. MBRS540T3G).

Calculated prediction is (substituting I_{load} with $\frac{V_{out}}{R}$):

$V_{out} = 4.73V$
$I_{load} = 3.79A$

Comparing with the simulation:

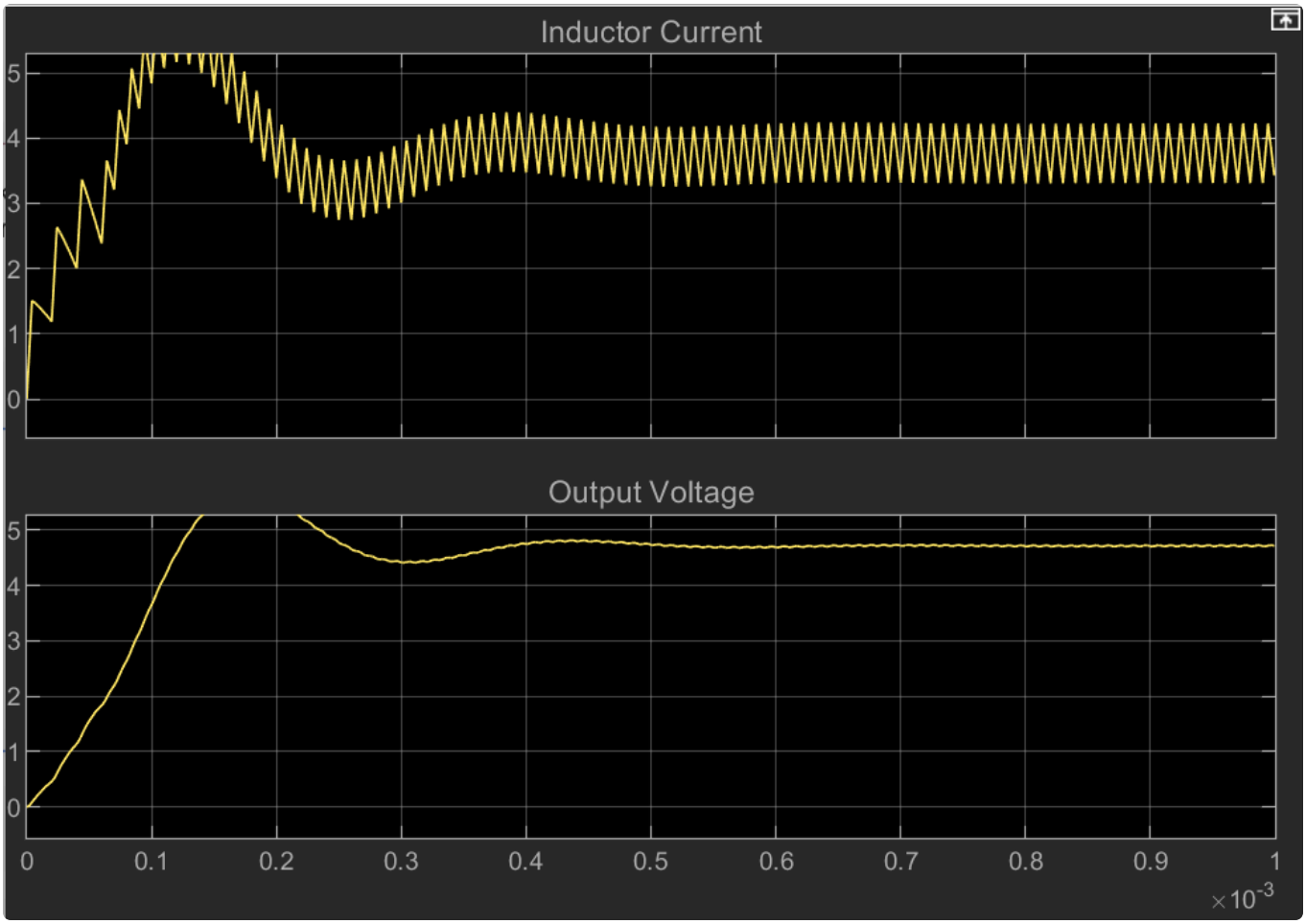


Figure 3. Inductor Current (A) and Output Voltage (V) readings for the chosen Schottky diode model in open-loop stage.

Results match almost perfectly, and it is a good improvement from earlier 4.07V.

To get even better results, a controller is needed.

Control Design